

# Calculus with Parametric and Polar Curves

ANSWER KEY



## Parametric derivatives and arc length

- 1 For the curve  $x = t^2$ ,  $y = t^3$ , find  $\frac{dy}{dx}$  in terms of  $t$ .

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2}{2t} = \frac{3t}{2} \text{ (for } t \neq 0\text{)}.$$

- 2 Find the arc length of the curve  $x = 3t^2$ ,  $y = 2t^3$  for  $0 \leq t \leq 2$ .

$$\frac{dx}{dt} = 6t, \frac{dy}{dt} = 6t^2. \\ L = \int_0^2 \sqrt{36t^2 + 36t^4} dt = \int_0^2 6t\sqrt{1+t^2} dt = [2(1+t^2)^{3/2}]_0^2 = 2(5^{3/2} - 1) = 10\sqrt{5} - 2.$$

## Polar area

- 3 Find the area enclosed by one loop of the rose  $r = 3\sin 2\theta$  (for  $0 \leq \theta \leq \pi/2$ ).

$$A = \frac{1}{2} \int_0^{\pi/2} 9\sin^2 2\theta d\theta = \frac{9}{2} \int_0^{\pi/2} \frac{1 - \cos 4\theta}{2} d\theta = \frac{9}{4} \left[ \theta - \frac{\sin 4\theta}{4} \right]_0^{\pi/2} = \frac{9\pi}{8}.$$

- 4 Find the area enclosed by the cardioid  $r = 2 + 2\cos \theta$ .

$$A = \frac{1}{2} \int_0^{2\pi} (2 + 2\cos \theta)^2 d\theta = \frac{1}{2} \int_0^{2\pi} (4 + 8\cos \theta + 4\cos^2 \theta) d\theta. \text{ The } \cos \theta \text{ term integrates to 0 over a full period, and } \int_0^{2\pi} 4\cos^2 \theta d\theta = 4\pi: A = \frac{1}{2}(8\pi + 4\pi) = 6\pi.$$

## Second derivatives and concavity for parametric curves

- 5 For  $x = t^2$ ,  $y = t^3 - 3t$ , find  $\frac{d^2y}{dx^2}$  in terms of  $t$ , using  $\frac{d^2y}{dx^2} = \left[ \frac{d}{dt} \left( \frac{dy}{dx} \right) \right] \div \frac{dx}{dt}$ .  
 $\frac{dy}{dx} = \frac{3t^2 - 3}{2t}$ . Differentiating with respect to  $t$ :  $\frac{d}{dt} \left( \frac{3t^2 - 3}{2t} \right) = \frac{(6t)(2t) - (3t^2 - 3)(2)}{4t^2} = \frac{12t^2 - 6t^2 + 6}{4t^2} = \frac{6t^2 + 6}{4t^2}$ . Dividing by  $\frac{dx}{dt} = 2t$ :  $\frac{d^2y}{dx^2} = \frac{6t^2 + 6}{8t^3} = \frac{3(t^2 + 1)}{4t^3}$ .

## Area between polar curves

- 6 Find the area inside the circle  $r = 3$  and outside the circle  $r = 2$  (an annular region).

$$A = \frac{1}{2} \int_0^{2\pi} (3^2 - 2^2) d\theta = \frac{1}{2}(5)(2\pi) = 5\pi.$$

## Horizontal and vertical tangents on parametric curves

- 7 For  $x = t^2 - 4$ ,  $y = t^3 - 3t$ , find the value(s) of  $t$  where the tangent line is horizontal ( $\frac{dy}{dt} = 0$ ,  $\frac{dx}{dt} \neq 0$ ).  
 $\frac{dy}{dt} = 3t^2 - 3 = 0$  gives  $t = \pm 1$ . Check  $\frac{dx}{dt} = 2t \neq 0$  at both: valid. Horizontal tangents at  $t = 1$  and  $t = -1$ .

- 8 For the same curve, find the value(s) of  $t$  where the tangent line is vertical ( $\frac{dx}{dt} = 0$ ,  $\frac{dy}{dt} \neq 0$ ).  
 $\frac{dx}{dt} = 2t = 0$  gives  $t = 0$ . Check  $\frac{dy}{dt} = 3(0)^2 - 3 = -3 \neq 0$ : valid. Vertical tangent at  $t = 0$ .

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## Converting between polar and Cartesian for calculus

- 9 Find the slope of the tangent line to  $r = 1 + \sin \theta$  at  $\theta = 0$ , using  $x = r \cos \theta$ ,  $y = r \sin \theta$  as parametric equations in  $\theta$ .

$x = (1 + \sin \theta) \cos \theta$ ,  $y = (1 + \sin \theta) \sin \theta$ .  $\frac{dx}{d\theta} = \cos \theta \cos \theta - (1 + \sin \theta) \sin \theta = \cos^2 \theta - \sin \theta - \sin^2 \theta$ ;  
at  $\theta = 0$ :  $1 - 0 - 0 = 1$ .  $\frac{dy}{d\theta} = \cos \theta \sin \theta + (1 + \sin \theta) \cos \theta$ ; at  $\theta = 0$ :  $0 + 1 = 1$ . Slope  $= \frac{dy/d\theta}{dx/d\theta} = \frac{1}{1} = 1$ .

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## Speed and length for parametric motion

- 10 A particle moves according to  $x(t) = t^2$ ,  $y(t) = 4t$ . Find its speed at  $t = 3$ .

$\frac{dx}{dt} = 2t$ ,  $\frac{dy}{dt} = 4$ . Speed  $= \sqrt{(2t)^2 + 4^2}$ ; at  $t = 3$ :  $\sqrt{36 + 16} = \sqrt{52} = 2\sqrt{13}$ .

- 11 Find the arc length of  $x = \cos^3 t$ ,  $y = \sin^3 t$  for  $0 \leq t \leq \frac{\pi}{2}$  (an astroid arc), using  $\frac{dx}{dt} = -3\cos^2 t \sin t$  and  $\frac{dy}{dt} = 3\sin^2 t \cos t$ .

$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9\cos^4 t \sin^2 t + 9\sin^4 t \cos^2 t = 9\sin^2 t \cos^2 t (\cos^2 t + \sin^2 t) = 9\sin^2 t \cos^2 t$ . So the speed is  $3|\sin t \cos t|$ .  $L = \int_0^{\pi/2} 3\sin t \cos t dt = \left[\frac{3}{2}\sin^2 t\right]_0^{\pi/2} = \frac{3}{2}$ .

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## Slope and concavity for polar-derived curves

- 12 Find  $\frac{dy}{dx}$  for the polar curve  $r = 2\theta$  at  $\theta = \frac{\pi}{2}$ , using  $x = r \cos \theta$ ,  $y = r \sin \theta$ .

$x = 2\theta \cos \theta$ ,  $y = 2\theta \sin \theta$ .  $\frac{dx}{d\theta} = 2\cos \theta - 2\theta \sin \theta$ ; at  $\theta = \frac{\pi}{2}$ :  $0 - \pi = -\pi$ .  $\frac{dy}{d\theta} = 2\sin \theta + 2\theta \cos \theta$ ; at  $\theta = \frac{\pi}{2}$ :  $2 + 0 = 2$ . Slope  $= \frac{2}{-\pi} = -\frac{2}{\pi}$ .

- 13 For  $x = t^3 - 3t$ ,  $y = t^2$ , find the points where the tangent is horizontal.

$\frac{dy}{dt} = 2t = 0$  gives  $t = 0$ . Check  $\frac{dx}{dt} = 3t^2 - 3 = -3 \neq 0$  at  $t = 0$ : valid. Horizontal tangent at  $t = 0$ , i.e. the point  $(x, y) = (0, 0)$ .

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## Arc length of a polar curve

- 14 Set up (do not evaluate) the arc length integral for the circle  $r = 4$  over  $0 \leq \theta \leq 2\pi$ , using

$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$ , and verify the setup is consistent with the known circumference formula.

$\frac{dr}{d\theta} = 0$ :  $L = \int_0^{2\pi} \sqrt{16 + 0} d\theta = \int_0^{2\pi} 4 d\theta = 8\pi$ , matching the circumference  $2\pi r = 2\pi(4) = 8\pi$  of a circle with radius 4.

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## Applied parametric motion

- 15 A projectile's position is given by  $x(t) = 20t$ ,  $y(t) = 15t - 4.9t^2$  (meters, seconds). Find the projectile's velocity components  $\frac{dx}{dt}$  and  $\frac{dy}{dt}$ , and find the speed at  $t = 1$ .

$$\frac{dx}{dt} = 20, \frac{dy}{dt} = 15 - 9.8t; \text{ at } t = 1: \frac{dy}{dt} = 5.2. \text{ Speed} \\ = \sqrt{20^2 + 5.2^2} = \sqrt{400 + 27.04} = \sqrt{427.04} \approx 20.66 \text{ m/s.}$$

- 16 For the same projectile, find the time at which the vertical velocity is 0 (the peak of the trajectory).

$$\frac{dy}{dt} = 15 - 9.8t = 0 \text{ gives } t = \frac{15}{9.8} \approx 1.53 \text{ s.}$$

## Polar curves and symmetry

- 17 The cardioid  $r = 1 + \cos \theta$  is symmetric about the polar axis. Use this symmetry to explain why the total area enclosed can be computed as  $2 \times$  the area for  $0 \leq \theta \leq \pi$ , and evaluate the total area.

Since  $r(\theta) = r(-\theta)$ , the curve is symmetric about the polar axis, so the area for  $\theta \in [0, \pi]$  equals the area for  $\theta \in [-\pi, 0]$ . Total area

$$= 2 \cdot \frac{1}{2} \int_0^\pi (1 + \cos \theta)^2 d\theta = \int_0^\pi (1 + 2\cos \theta + \cos^2 \theta) d\theta = \left[ \theta + 2\sin \theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^\pi = \pi + 0 + \frac{\pi}{2} + 0 = \frac{3\pi}{2}$$

- 18 For the parametric curve  $x = e^t \cos t$ ,  $y = e^t \sin t$ , find  $\frac{dy}{dx}$  at  $t = 0$ , using  $\frac{dx}{dt} = e^t(\cos t - \sin t)$  and  $\frac{dy}{dt} = e^t(\sin t + \cos t)$ .

$$\text{At } t = 0: \frac{dx}{dt} = e^0(1 - 0) = 1. \frac{dy}{dt} = e^0(0 + 1) = 1. \text{ Slope } = \frac{dy/dt}{dx/dt} = \frac{1}{1} = 1.$$

- 19 A Ferris wheel of radius 10 m, centered 12 m above the ground, rotates so that a rider's position is  $x(t) = 10\cos t$ ,  $y(t) = 12 + 10\sin t$ , where  $t$  is measured in radians and represents elapsed angle since the rider was at the rightmost point. Find the rider's height above the ground when  $t = \frac{\pi}{3}$ , and find the vertical velocity  $\frac{dy}{dt}$  at that instant (assuming  $t$  increases at a constant rate of 1 rad per unit time).

$$\text{Height: } y\left(\frac{\pi}{3}\right) = 12 + 10\sin\left(\frac{\pi}{3}\right) = 12 + 10 \cdot \frac{\sqrt{3}}{2} = 12 + 5\sqrt{3} \approx 20.66 \text{ m. Vertical velocity: } \frac{dy}{dt} = 10\cos t; \text{ at } t = \frac{\pi}{3}: 10 \cdot \frac{1}{2} = 5 \text{ (units of height per unit time).}$$

## Total distance traveled along a parametric path

- 20 A robot on a factory floor follows the path  $x(t) = t^2$ ,  $y(t) = 2t$  for  $0 \leq t \leq 3$  (meters, seconds). Set up the arc length integral for the total distance traveled, and evaluate it exactly using the substitution  $u = t^2 + 1$  where appropriate.

$$\frac{dx}{dt} = 2t, \frac{dy}{dt} = 2: L = \int_0^3 \sqrt{4t^2 + 4} dt = 2 \int_0^3 \sqrt{t^2 + 1} dt. \text{ This does not simplify via a simple power-rule substitution (it requires a trigonometric or hyperbolic substitution to evaluate in closed form), but can be set up as } L = 2 \int_0^3 \sqrt{t^2 + 1} dt, \text{ which numerically evaluates to approximately } 2(5.20) = 10.40 \text{ m.}$$